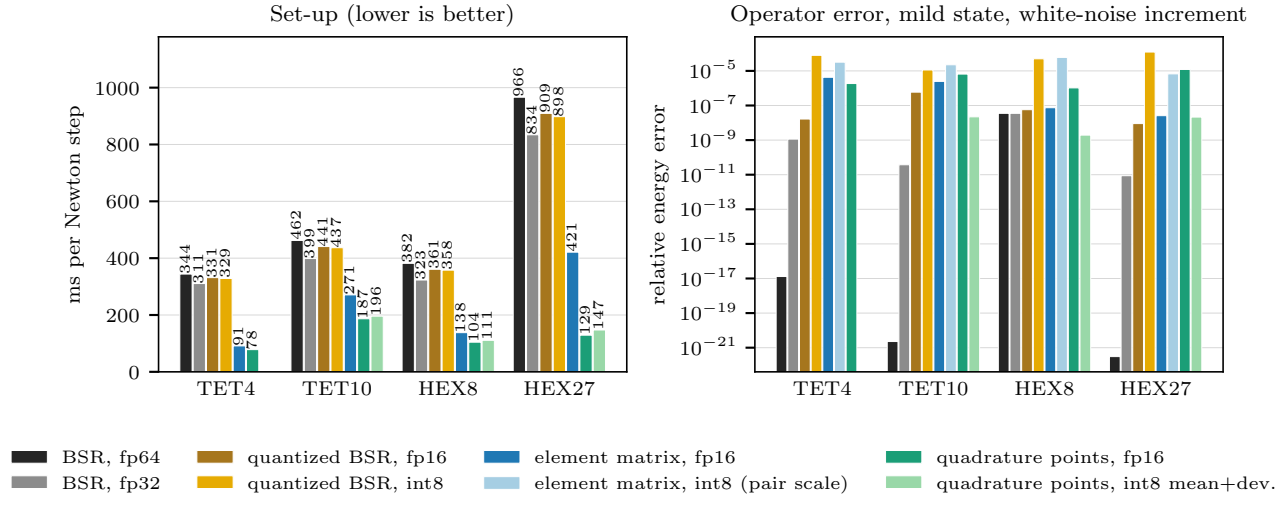

Set-up (lower is better)
ms per Newton step
0
200
400
600
800
1000
344
311
331
329
91
78
462
399
441
437
271
187
196
382
323
361
358
138
104
111
966
834
909
898
421
129
147
TET4
TET10
HEX8
HEX27
Operator error, mild state, white-noise increment
relative energy error
10^{-5}
10^{-7}
10^{-9}
10^{-11}
10^{-13}
10^{-15}
10^{-17}
10^{-19}
10^{-21}
TET4
TET10
HEX8
HEX27
BSR, fp64
BSR, fp32
quantized BSR, fp16
quantized BSR, int8
element matrix, fp16
element matrix, int8 (pair scale)
quadrature points, fp16
quadrature points, int8 mean+dev.